Course: **Explainable and Responsible AI (XRAI)**

Topic: **AI Model Governance Framework**

Reference: **Companion to the Model AI Governance Framework – Implementation and Self-Assessment Guide for Organizations**

https://www.pdpc.gov.sg/-/media/files/pdpc/pdf-files/resource-for-organisation/ai/sgisago.pdf

Section 2:

Internal governance structures and measures

To guide organizations to develop appropriate internal governance structures

Guiding Questions	Useful industry examples, practices and guides for consideration	Remarks

2.1 Does your organization have an existing governance structure that can be leveraged to oversee the organization's use of AI?
 	
2.2 If your organization does not have an existing structure to tap on, has your organization put in place a governance structure to oversee the organization's use of AI?
	- To adapt existing Governance, Risk and Compliance (GRC)
	
	To provide oversight on the use of data and AI within an organisation:-
	
	- Sandbox type of governance to test-bed and deploy AI solutions
	  o prior to fully fledged governance structures
	- committee with representatives from
	  o relevant department
	  ▪ legal/compliance, technical and sales and communication
	  o with proper terms of references
	  ▪ refine organization's AI governance
		Governance
		- Corporate governance
		- IT governance
		- Data governance
		- Analytics governance
		- AI governance
		
		- Data Governance Office (DGO)
		
		- Cross- functional committee
		
		- Change Control Board (CCB) / Change Authority Board (CAB)
		
		- AI Ethics Review Board
		  o A diverse group of stakeholders and expertise

	frameworks to ensure they meet the organization's commercial, legal, ethical and reputational requirements
	- subject matter experts
	  o chief security officer
	  o data protection officer
	- checks and balances
	  o oversee methodology, algorithm, deployment of AI
	  o conduct validation
	
	Implement the governance structure:-
	- decision–making approach
	  o **centralised**
	  ▪ AI solution not low risk issue
	  • Senior management
	  • AI ethics committee
	  o **decentralised**
	  ▪ deploy AI solution (whitelist vs blacklist)
	- review/recalibrate AI Models
	  o resources-efficient
	- regular reviews of governance processes and structures
		- **Ethics Committees**
		  o Centralised
		  o Decentralised

2.3 Did your organization's board and/or senior management sponsor, support and participate in your organization's AI governance?	- Committee/board
	  o Chaired by
	  ▪ Senior management
	  o Participated by
	  ▪ Senior leaders
	  • CDA, CPO, CISO…
		Traditional corporate/business functions vs digital excellence
		- **CxO**
		
		- Information
		- Technology
		- Finance
		- Operating
		- Customer
		- People/HR
		- **Data**
		- **Privacy**
		- **Security**
		- **Knowledge**

		- **Analytics** - **AI** - **AI Ethics (CAEO)** - **Ethics & Compliance (CECO)** - **Digital Transformation** **Mandate/ authority** - Top-down approach
Clear roles and responsibilities for the ethical deployment of AI		

2.4 Are the responsibilities of the personnel involved in the various AI governance processes clearly defined?	- Board/senior management to champion responsible AI deployment??
	  o **Strategic** level
	  ▪ Oversight
	  • Responsible risk and corporate values
	  • To approve AI models
	  o **Implementation** level
	  ▪ Project team leads and officers
	  • Responsible/ accountable for AI project
	  • Roles and responsibilities
	  o Manage model risk
	  o regulatory compliance
	  o documentation
	
	- separate **responsibilities**
	  o **business** staff
	  ▪ define business goals, business rules, AI systems behaviour
	  o **technical** staff
		Organisational structure/ matrix
		- strategic
		- tactical
		- operational
		
		- functional area/business domain

	▪ data practices, security, stability, error handling - review of job descriptions o roles involve AI deployment	
2.5 Are the personnel involved in various AI governance processes: A. Fully aware of their **roles** and **responsibilities**? B. Properly **trained**? C. Equipped with the necessary **resources** and **guidance** to perform their duties? 2.6 Are the relevant staff dealing with AI systems properly **trained** to **interpret AI model output and decisions** as well as to **detect and manage bias in data**? 2.7 Are the other staff who interact with the AI system aware of and sensitive to the relevant risks when using AI? Do they know who to raise such issues to when they spot them (e.g. subject-matter experts within their organizations)?	- Hiring talent o Right skillsets - Multi-disciplinary team o Impact of AI deployment on the organization and individuals - New and specialized role (eg: data scientist) o Specific responsibilities ▪ to examine ethical and data protection issues in AI deployment process - awareness for key internal stakeholders o implications of AI development/deployment o guidelines (eg: AI Engineering guidelines) - **training** o **general** training for personnel involved in various AI governance processes o **specialised** training for staff dealing with AI systems - develop and partner with education institute o online **learning** modules to support AI skill development for employees - **educate** employees at all levels o use AI o customer-facing roles ▪ to identify and report potential ethical concerns wrt AI development and deployment	-
Risk management and internal controls		

2.8 Does your organization have an existing risk management system that can be expanded to include AI-related risks?
 	
2.9 Did your organization implement a risk management system to address risks involved in deploying the identified AI solution (e.g. personnel risk or changes to commercial objectives)?
	- Implement
	  o Internal **policy** explanation process
	- Implement
	  o **Knowledge management registry** to archive relevant documents to ensure proper knowledge transfer
		-

Section 3:

Determining the level of human involvement in AI-augmented decision-making

To help organizations determine the appropriate extent of human oversight in their AI-augmented decision-making process

Guiding Questions	Useful industry examples, practices and guides for consideration	Remarks

3.1 Did your organization conduct an impact assessment (e.g. probability and/or severity of harm) on individuals and organizations who are affected by the AI solution?	- internal vs external stakeholders
	- assess risk
	  o technical perspective
	  ▪ eg: system integrity test
	  o personal data protection perspective
	  ▪ eg: PDPC's guide to Data Protection Impact assessment	

3.2 Based on the assessment, did your organization implement the appropriate level of human involvement in AI-**augmented** decision-making?	- Human-centric Design
	  o **Human-in-the-loop (HITL)**
	  ▪ Quality of the decision made
	  ▪ Human subjective judgement is required
	  o **Human-out-of-the-loop (HOOTL)**
	  ▪ Not practical to subject every algorithmic recommendation to human review
	  o **Human-over-the-loop (HOTL)**
	  ▪ To intervene when the situation calls for it
	  • Statistical confidence level (eg: threshold)
	- Factors in determining the level of human involvement
	  o Risk appetite
	  o User experience of its clients' customers
	  o Operational cost
		Collaboration of **Human-Machine** decision making process
		- from human to autonomous
		- 7 Levels to Blend Human and AI in Hybrid Decision Making
		- **Support → augmentation → automation**
		
		HITL
		- Life threatening environment
		  o Eg: chemical plant, hospital

3.3 After deployment, did your organization continually identify, review and mitigate risks of using the identified AI solution?	- **Regular review period** for retraining the AI model
	- Key Performance Indicators (**KPI**) for AI model's performance
	- Data drift
	  o Development/Training Data vs Production Data
	- **Scenario-based** response
	  o If risk management effort fail
		Regular review period
		- Data drift
		  o Change of input data (upstream via data lineage) and impact model (downstream)
		- Model drift
		  o Change of business requirement
		
		KPI
		- Prediction
		- Classification (confusion matrix – Accuracy, Precision, Recall, F1 Score,…)
		- Regression (Mean Squared Error – MSE, Root Mean Squared Error – RMSE)
		- Area Under the Curve (AUC)
		- Receiver Operating Characteristic (ROC)
		- R-squared
		- Training Time, Inference Time
		- Model Size
		- Overfitting/Underfitting
		- Feature Importance
		- etc
		
		Scenario analysis and modelling
		- Prediction/forecast
		  o Choice-based (radio button)
		- **Scenarios**/radars
		  o Option-based strategies (checkbox)

Relevant only in limited scenarios:	- System failure o Shutdown of AI system/ bring it back to safe state	**Human to review** - **HITL**

3.4 For safety-critical systems, did your organization ensure that:
 	
A. The relevant personnel will be able to assume control where necessary?
 	
B. The AI solution provides sufficient information to assist the personnel to make an informed decision and take actions accordingly?
	- AI model making decision – significantly unsure of answers/prediction
	  o Flag the cases
	  o Human to review
		o Exception handling

Section 4:

Operations Management

To help organisations adopt responsible measures in the operations aspect of their AI adoption process

Guiding Questions	Useful industry examples, practices and guides for consideration	Remarks
Data for Model Development - Ensuring personal data protection		
4.1 Did your organization implement accountability-based practices in **data management and protection** (e.g. the PDPA and OECD Privacy Principles)?	**Data governance** - Data **confidentiality**, **privacy**, **security** (data **masking** & data ***anonymization*** for ***pseudonymized*** or de-identified data) **Personal Data Protection Act (PDPA)** - ***2nd lecture slides (references)*** - PDPC's Data Protection Trustmark - https://www.pdpc.gov.sg/overview-of-pdpa/data-protection/business-owner/data-protection-trustmark - Data Lifecycle - https://www.pdpc.gov.sg/-/media/Files/PDPC/PDF-Files/Other-Guides/DPMP/Data-Flow-Illustration-PDF-v2.pdf - Data Protection Notice Generator - https://apps.pdpc.gov.sg/dp-notice-generator/ - Data Protection Obligations - https://www.pdpc.gov.sg/overview-of-pdpa/the-legislation/personal-data-protection-act/data-protection-obligations **Asia Pacific Economic Cooperation Cross Border Privacy Rules and Privacy Recognition for Processors (APEC CBPR & PRP) Systems certifications**	**Data monetisation** **Data sharing** - Must share data unless proven otherwise **Data minimisation** - Data collection vs data connection **Data encryption** **Data poisoning** - attacks an AI system by injecting bad data into AI model's training data set **Data enrichment** **AI ready data** - Synthetic data - Artificial generated data **Data protection** - Data protection Impact Assessment (DPIA) - Data Protection Officer (DPO) o Expert on data protection law

	- APEC Cross Border Privacy Rules (CBPR) - https://www.imda.gov.sg/how-we-can-help/cross-border-privacy-rules-certification - APEC Privacy Recognition for Processors (PRP) System - https://www.imda.gov.sg/how-we-can-help/privacy-recognition-for-processors-certification **Organisation for Economic Co-operation and Development (OECD)** - The OECD Privacy Framework o https://www.oecd.org/sti/ieconomy/oecd_privacy_framework.pdf Data Governance Panel/ Dashboard - To assist **Governance, Risk and Compliance (GRC)** for **Data Protection**	
Data for Model Development - Understanding the lineage of data		
4.2 Did your organization implement measures to trace the **lineage** of data (i.e. backward data lineage, forward data lineage and end-to-end data lineage)?	- Data provenance - Data inventory, **data dictionary**, data change processes, document control mechanisms - Data policy team - **Governance** team o **Data** governance, **analytics** governance, **AI** governance	**Data dictionary** - Capture various metadata
Relevant only in limited scenarios: 4.3 If your organization obtained datasets from a **third party**, did your organization assess and manage the risks of using such datasets?	- Data protection - IMDA's Trusted Data Sharing Framework o https://www.imda.gov.sg/-/media/Imda/Files/Programme/AI-Data-Innovation/Trusted-Data-Sharing-Framework.pdf	-

Data for Model Development - Ensuring data quality		
4.4 Is your organization able to verify the accuracy of the dataset in terms of how well the values in the dataset match the true characteristics of the entity described by the dataset?	Metadata Data categorisation using Data Taxonomy - Taxonomy of **data annotation** to standardise the process of **data labelling**	**Data lifecycle** - Data preparation - Data profiling and inspection (particularly for data quality concern) **Dimension** - Data source - Data domain - Data category Data Taxonomy Example 1 using **Data Source** dimension: - Authored data - User-entered data - Captured data - Derived data Data Taxonomy Example 2 using **Data domain** dimension: - HR – PII - Finance & Payroll - Healthcare – patient details Data Taxonomy Example 3 using **Data Category** dimension: - Master data - Reference data - Metadata - Unstructured data - Transactional data (OLTP) - Historical data (DW, OLAP, etc)
4.5 Is the **dataset** used complete in terms of **attributes** and items? 4.6 Is the dataset used credible and from a reliable **source**?	Data schema Exploratory data analysis (prior to testing) o Error o Inconsistencies Data monitoring and reporting processes	Data schema - Data modelling - Entity relationship modelling vs - dimensional modelling (star schema, snowflake schema, etc)

4.7 Is the dataset used **up-to-date**? 4.8 Is the dataset used **relevant**? 4.9 Where **personal data** is involved, is it collected for the intended purposes? 4.10 Is the dataset used well-structured and in a machine-understandable form? *Relevant only in limited scenarios:* 4.11 If the dataset used has been joined from multiple datasets, were the extraction, transformation and other relevant operations performed correctly?	**Data classification** principles o Legal and data governance frameworks and standards ▪ International Organization for Standardization (ISO) guidelines Data engineering team - AI solution and business needs - ETL process - Denormalise or transform - Unit test - Upstream data source changes o Downstream model impact	Data ecosystem - data fabric - data mesh - data hub *Note:* *enrich the data, don't just classify the data.* Normalisation (OLTP) vs denormalisation (historical – Logical DW, DW, Data Lake, Data Lakehouse, etc) *Note:* *Normalisation in Statistics vs Normalisation in Database* Upstream + downstream data flow - Data lineage - Data mapping
4.12 If any human has filtered, applied labels, or edited the data, did your organization implement measures to ensure the quality of dataset used?	Data pipepline - Data role – accountability - Who manipulated data, by which rule	Data Personnel - Data owner/ producer - Data manager - Data custodian - Data steward Data consumer - Data engineer - Data analyst - Data scientist - AI Engineer
Data for Model Development - Minimising inherent bias		
4.13 Did your organization take steps to mitigate	Personally Identifiable Information (PII) - Social / demographic data - Sensitive/ protected attributes	Data minimisation - Data collection (collecting data and

unintended **biases** in the dataset used for the AI model, especially omission bias and stereotype bias?
 	
4.14 Did your organization use a complete dataset by not removing data attributes prematurely to minimize risk of inherent bias?
 	
Relevant only in limited scenarios:
 	
4.15 Did your organization take steps to mitigate biases that may result from data collection devices (e.g. cameras and sensors)?
	Bias of data annotation
	
	Data attribute
	
	Statistical tool
	- To evaluate bias
		to be stored internally)
		vs
		- Data connection (connecting to external data source for real-time consumption)
		
		Data collection and storage
		- Cost of storage (getting cheaper!)
		- Downstream impact on data governance
		
		Data bias
		- Avoid cherry-picking
		- Synthetic data
		
		- Power BI
		  o visual inspection
		- Data profiling in Power Query Editor
		- R/Python
		  o Exploratory Data Analysis

4.16 Is the dataset used to produce the AI model fully representative of the actual data or environment the AI model may receive or function in?	*To mitigate selection bias, consider:-* Data representative - Sample Data distribution vs Population Statistics o Identify & quantify Data quality metrics - Completeness - Freshness - Context (context before content) Heterogeneous dataset - Data collected from different demographic groups	

Data for Model Development - Different datasets for training, testing and validation		
4.17 Did your organization use different datasets for training, testing and validation of the AI model?	Training vs validating Statistical test - eg: Receiver Operating Characteristic Curve (ROC)	training dataset - for model to learn from the training dataset and directly improve its parameters validation dataset - to evaluate and fine-tune a model during training - to assess the model's performance and make adjustments test dataset - to evaluate the final performance of a trained model
4.18 Did your organization test the AI model used on different demographic groups to mitigate systematic bias? *Relevant only in limited scenarios:* 4.19 Did your organization split a large dataset into subsets to mitigate risks of systematic bias when validating the AI model?	Data drift	-
Data for Model Development - Periodic review and updating of datasets		
4.20 Did your organization **periodically review** and update datasets to ensure its accuracy, quality, currency, relevance and reliability?	*To ensure **data accuracy**, **quality**, **currency**, **relevance**, and **reliability**, consider:-* Schedule regular review of dataset New data availability	

4.21 Did your organization implement measures to minimize reinforcement bias?	Data bias → Model bias Consider prioritizing - **Explainability** - **Robustness** - **Regular tuning**	
Algorithm and Model		

4.22 Did your organization identify relevant features or functionalities that have the greatest impact on your organization's stakeholders?
 	
4.23 Did your organization identify which measures will be the most effective in building trust with your organization's stakeholders?
	Prioritization
	- Top priority
	  o **Explainability**
	  o **Robustness**
	  o **Regular tuning**
	- Secondary priority (resource-intensive, relevant for specific purposes – accreditation & certification)
	  o **Traceability**
	  o **Reproducibility**
	  o **Auditability**
		-

Algorithm and Model - Explainability		
4.24 Can your organization explain how the deployed AI model functions and arrives at a particular prediction?	AI Model operates AIOps - MLOps, ModelOps, DataOps, DevOps Simpler models vs complex - eg: linear regression vs NN/ML/DL Use visualisation - eg: graph Use narrative - eg: correlation between factors, simple indicators – high/medium/low Documentation for AI Modelling To explain significant feature contributing to the output	White-box vs black-box

	- Eg: LIME, SHAP	
Algorithm and Model - Repeatability		
4.25 Where explainability cannot be practically achieved, did your organization consider lesser alternatives?	Test and validate Conduct Simulation - To collect & correlate data from different ecosystems for quality control and ensure real-world validation Implement version control (AI models)	Simulation (descriptive/ predictive/ prescriptive analytics) - Monte Carlo simulation - Pseudo-randomisation via random seed number Versioning control - Provisioning of codes to various environment - Testing, UAT, Pre-Production, Production
Algorithm and Model - Robustness		
4.26 Did your organization ensure that AI model deployed is sufficiently robust?	Designing, verifying and validating AI models Adversarial testing - Unexpected input variables (eg: unexpected changes or anomalies)	-
Algorithm and Model - Active monitoring, review and tuning		
4.27 Did your organization perform active monitoring, review and regular model tuning when appropriate (e.g. changes to customer behaviour, commercial objectives, risks and corporate values)?	*Where practical and/or relevant, consider:* New data points - Data pipeline o ETL - Retrain model User feedback	Generative AI - Reinforcement Learning with Human Feedback (RLHF) - Reinforcement Learning with AI Feedback (RLAIF)
4.28 Did the AI model testing reflect the actual production environment it is	*Where practical and/or relevant, consider:* Create test cases & run model scenarios	POC - prototype

supposed to operate in?	- Eg: what-if Run proof-of-concept	
4.29 Did your organization assess the degree to which the identified AI solution generalized well and failed gracefully?	Create automated tool *To assess whether the AI solution failed gracefully, consider:* **Accountability** - Use confidence levels & threshold Communication with stakeholders Business continuity plan	-
Algorithm and Model - Traceability		
Relevant only in limited scenarios: 4.30 Did your organization document the relevant information such as datasets and processes that yield the AI models' decisions in an easily understandable manner?	*Where practical and/or relevant, consider:* Track AI model's decision-making process & performance o Standard documentation/ dashboard a) Project objectives b) Use of data and input values (eg: data dictionary) c) Research approach d) Algorithm description, evaluation parameters & AI model outcomes e) Error logs & error rate metrics (eg: false acceptance rate & throughput metrics) f) Notes by technical team (eg: technical writing) Audit trail Data - Keep a copy of training data & document how data was processed	
Algorithm and Model - Reproducibility		
Relevant only in limited scenarios: 4.31 Did your organization engage an independent team to check if they	*Where practical and/or relevant, consider:* Take into account specific contexts or particular conditions	Eg: Monte-carlo simulation – to use random seed Pseudorandom – to reproduce same output for multiple runs

can produce the same or very similar results using the same AI method based on the documentation relating to the model made by your organization?	Replicate files for testing and reproducing	
Algorithm and Model - Auditability		
Relevant only in limited scenarios: 4.32 Has your organization put in place relevant documentation, procedures and processes that facilitate internal and external assessments of the AI system?	*Where practical and/or relevant, consider:* Evaluation by internal vs external assessors Comprehensive record - Data provenance, procurement, pre-processing, data lineage, data storage and data security Centralise information digitally in process log	Process log - audit trail security concern - identity & access management (IAM) (**AAA**) a) Authentication b) Authorisation c) Accountability/ Auditing